LISP – Senior Division

1. **01-02 C1 Lisp Programming**

   Evaluate: (CAR'((2 (3)) (4 (5 6)) 7))

2. **02-03 C1 Lisp Programming**

   What is the value of the following LISP program?

   ```
   (SET Q CDS '(J (Q K) A))
   (SET Q OTH '(CONS '10 CDS))
   (REVERSE (CAR (CDR (CDR OTH))))
   ```

3. **03-04 C1 Lisp Programming**

   Evaluate: (SQUARE (DIV (EXP 2 3) (SUB 10 8)))

4. **04-05 C1 Lisp Programming**

   Evaluate: (CAR (CDR (REVERSE (CDR' ((A B) C (D E F) (G H))))))

5. **05-06 C1 Lisp Programming**

   Evaluate: (CDR (CDR (CAR (CDR '(3 (4 1 (3 2)) 1 (2 (1 3) 4))))))

6. **06-07 C1 Lisp Programming**

   Evaluate: (CDR (CAR (CDR (CDR' (2 (3 1) (4 5 (2 3)) (6 1 (8 2)))))))

7. **07-08 C1 Lisp Programming**

   Evaluate the following: (CDR (CAR (CDR (REVERSE (CDR '((2 3) (1 2) ((2 3) 1) 2))))))

8. **08-09 C1 Lisp Programming**

   Evaluate the following: (CAR (REVERSE (CDR '(3 2 (4 (1 2)) 7 6 (2 1)))))

9. **09-10 C1 Lisp Programming**

   Evaluate: (CDR (CAR (REVERSE (CDR '((8 2) 3 ((8 1) 7) 2 (1 (2 3)))))))

10. **10-11 C1 Lisp Programming**

    Evaluate: (ADD (MULT 2 3) (EXP 3 2) (SUB 5 2) (DIV 8 2))

11. **11-12 C2 Lisp Programming**

    Evaluate the following expression: (EXP 2 (DIV (MULT 8 6) (SQUARE 4)))

12. **11-12 C4 Lisp Programming**

    Evaluate the following expression: (CAR (CDR (CAR (CDR '((1 2) (3 (4 5)) 4 (3 2) 1))))))

13. **12-13 C2 Lisp Programming**

    Evaluate the following expression: (SUB (ADD 2 (MULT 5 2) 3) (DIV 9 (EXP 3 1)))

14. **12-13 C4 Lisp Programming**

    Simplify the following expression:

    (CAR (CAR (CDR (REVERSE (CDR '((2 (1 2)) (4 1 3) ((2 1) 3) 4))))))

15. **13-14 C2 Lisp Programming**

    Evaluate the following expression: (MULT (ADD (SUB 10 2) 3) (EXP 2 4))

16. **13-14 C4 Lisp Programming**

    Evaluate: (CAR (CAR (REVERSE '((A B) C ((D E) F)))))

17. **14-15 C2 Lisp Programming**

    Evaluate the following LISP expression:

    (ADD (MULT 3 (EXP 2 3)) (MULT (EXP 2 2) (SUB (MULT 5 3) 6)) (EXP 2 (SUB (9 3))))

18. **14-15 C4 Lisp Programming**

    Evaluate the following LISP expression:

    (REVERSE (CAR (CDR (CDR (CAR '((a b (c d)) (e (f g)) (h (j k)))))))

19. **15-16 C2 Lisp Programming**

    Evaluate the following LISP expression:

    (ADD (MULT 2 5) (EXP 3 2) (SUB 9 (ADD 1 7)) (DIV 15 3) (ADD 4 (SUB 8 4) 6))

20. **15-16 C4 Lisp Programming**

    Simplify the following LISP expression:

    (CDR (CAR (CDR (REVERSE (CAR (CDR '(1 (2 ((3 4) 2) 3) 2 (4 (1 2)))))))))